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The Open Empirical Cycle for Hypothesis Evaluation in Psychology

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Abstract

In the last decade it has become clear that replicability of empirical psychological research should be better. *Open science* practices aim to enhance the transparency and accessibility of research thereby both enabling others to reproduce the results presented in a paper and increasing the replicability of these results. Examples of these practices include preregistration, publication of data and analyses, open access publications, and replication research. Although open science practices are gaining traction, they have rarely been placed in a broader epistemological context. To address this shortcoming, this paper introduces the *open empirical cycle*. It draws upon De Groot's empirical cycle, a model of cumulative knowledge generation via scientific research and adds open science practices. Adhering to the open empirical cycle, if only partly, will structure the scientific workflow and create awareness of the adverse consequences of deviations. Following the open empirical cycle increases the transparency, quality, trustworthiness, and replicability of research.

Keywords: Empirical Cycle, FAIR Data, Open Access, Preregistration, Replication Research

The Open Empirical Cycle for Hypothesis Evaluation in Psychology

Introduction

This paper introduces the *open empirical cycle*, in which open science practices are integrated in De Groot's empirical cycle (1961). De Groot's empirical cycle is a model of cumulative knowledge acquisition through scientific research. It consists of a repeating sequence of five steps: observation of phenomena to be explained; induction of an explanatory theory; deduction of testable predictions, that is, in this paper, hypotheses deduced from a theory; testing these predictions in empirical data; and evaluation, that is, reflecting back on the theory based on the results of the study that has been executed. As is highlighted in Figure 1, De Groot's cycle becomes *open* by integrating the open science practices replication, preregistration or registered reports, FAIR data and analyses, and open access publication. As will be elaborated in this paper, an important feature of the open empirical cycle is that, in the first three steps, research questions are formulated, data collection and analyses plans are determined, and published in the form of a preregistration or a registered report. These steps can be called prehoc, that is, executed before the data are collected. The last two steps are posthoc and concern the execution of the plans formulated in the first three steps.

In several fields of science the replicability of published research is rather poor (e.g., see reproduciblity project psychology[1], Open Science Collaboration, 2012, 2015, and reproducibilty project cancer research, Errington et al., 2021b). The empirical cycle can help to diagnose the current meta-scientific replication crises in these fields. As was already alluded to by De Groot (see de Groot et al., 2014, Annotations to Section 1), deviations from the cyclical process of the empirical cycle risk introducing questionable research practices (QRPs, Wicherts et al., 2016). QRPs increase the probability of false positive results, thereby undermining replicability. One example of deviation from the empirical cycle is moving backwards from testing in the fourth step to deduction in the third step.

[1] https://osf.io/ezcuj/

This introduces the QRP known as "hypothesizing after the results are known". This happens, for example, when researchers analyze data without an explicit hypothesis (for example by adding or removing covariates) until they find an interesting (or significant) effect, and then write a paper that presents this effect as if it were a test of an a-priori specified hypothesis. Another deviation is moving backwards from testing in the fourth step to induction in the second step. For example, after observing that a scale score only rendered insignificant results, researchers may speculate that sub scales of the total scale will show the expected effect. These phenomena were already described by Ioannides (2005) in his seminal paper "Why most published research findings are wrong", and can be framed in the context of the (open) empirical cycle (Wagenmakers et al., 2018). These deviations may occur because researchers are unaware of the consequences, adhere to the research culture in their group, or because of opportunistic behavior. It is our hope that increased awareness of the open empirical cycle will help researchers avoid deviations, or at least notice and disclose necessary deviations. Doing so will help to avoid QRPs, thereby making research less error prone and thus increasing its quality and replicability (Gopalakrishna et al., 2021; Makel, Hodges, & Cook 2021; Xie, Wang, & Kong, 2021).

What furthermore became clear from the reproducibility projects psychology and cancer research is that a lack of transparency by the authors of the original studies often makes it difficult to impossible to design and execute replication studies (Errington et al., 2021a). It may, for example, be that the data from the original study are unavailable or insufficiently documented, that analysis codes are lacking, that descriptives, effect sizes, and/or p-values are not reported, or that information about experimental setup, coding schemes or questionaires is lacking. While de Groot's empirical cycle does provide an epistomological basis for research, it does not address the transparency of research. Transparency is addressed by integrating open science practices such as replication studies, preregistration or registered reports, FAIR data and analyses, and open access publications into de Groot's empirical cycle. Public distrust in science (Broek-Honingh & de Jonge,

2018) can be alleviated by increasing transparancy (Rathenau Instituut, 2021). Distrust in science may cause significant harm to individuals and societies when scientific evidence is neglected. For example, politicians and policy makers may not receive the support for strategies that could clearly mitigate climate change, and individuals may choose not to use safe vaccines for themselves or their children.

This paper will continue with a discussion of De Groot's empirical cycle. Each step will be elaborated followed by a discussion of the open science practice belonging to that step. The paper will conclude with a discussion highlighting implications of the open empirical cycle not for science (that will be covered in the paper) but for scientists.

The open empirical cycle

Step 1: Observation of phenomena to be explained, that is, prepare the research project

In this step any of a number of activities can be executed: reconsideration of previous research results; studying and summarizing existing literature addressing the research area of interest; (re)analysis of existing data; evaluation and (re)formulation of theories; and naturalistic observations. The resulting summary of the state of affairs in the research area of interest can subsequently be augmened with the researcher's observation of gaps in the literature, additional variables that should be considered, or questions with respect to the validity or generalizability of research outcomes.

It is tempting to keep a mental record of the summary and to move on to the next step. However, actually writing up the literature review and theory sections of the envisaged paper and creating a reference list will enforce clarity and will be very useful when registering the research plans in Step 3. One of the outcomes of this step are verbal descriptions of the variables that are relevant to explore the research area of interest. Another outcome are the research questions, expressed in terms of the relevant variables, accompanied by an argument why it adds to the current knowledge, and will be impactful

and relevant.

At least three different situations can be distinguished. First and rather common, is the study of research questions using existing knowledge as the point of departure. These studies have a confirmatory nature, to formulate it in the Groot's terms: the "observations" collected in this step lead to "induced" expectations and theories in Step 2, from which in Step 3 testable consequences will be "deduced". Second and gaining importance is the execution of a replication study (Nosek & Errington, 2020) in order to verify the outcomes of an earlier "original" study. Replications can be direct, that is, copying the original study as closely as possible. The goal can be to determine whether two studies agree or are in conflict with each other, or, to meta-analytically summarize the results of the two studies into one overall conclusion. Direct replications can be extended by adding a new feature, e.g., by considering an additional experimental group, other populations, and/or additional confounders. Replications can also be conceptual, that is, compared to the "original" study, designed with new experimental manipulations, different operationalizations of variables, etc, but, still investigating the same underlying research question. Third, research areas may also, as to yet, be (relatively) unexplored. In these cases exploratory studies can be executed to obtain a first impression of a research area and generation of explananda, that is, phenomena that have to be explained. For a discussion of exploratory research in the context of the empirical cycle, see Van Lissa (2022a) and Höfler et al. (2023).

Step 2: Induction of an explanatory theory, that is, formulate expectations and theory

Based on the observations from Step 1, in this step a theory is induced and expectations (possibly competing) with respect to the answers to the research questions are formulated. In a survey, the expectation could be that prediction of an outcome (e.g., being infected with covid or not) improves if variables characterizing a persons social network are used in addition to demographic variables like gender, age, and income. In an

experiment the expectation could be that "the improvement" is smallest in the control group larger in the traditional treatment group and largest in the new treatment group. The competing expectation could be that this is not the case. The interested reader is referred to Fried (2020), who thoughtfully reflects on theory formulation.

Again, it is tempting to keep a mental record of the (lack of) expectations and theory and to move on to the next step. However, writing them down will again enforce clarity and will be very useful in the next step. After Step 2 it is possible to return to Step 1 (see Figure 1), re-evaluate the literature, read additional literature, and/or re-think before re-entering Step 2 once more.

Step 3: Deduction of testable predictions, that is, formulate hypotheses, plan data collection and analysis

A very short characterization of deduction in terms of de Groot's empirical cycle is "deriving testable consequences from the theory that has been formulated" . Testable consequences, that is in this paper, statistical hypotheses, are not formulated in a void, but in a context that clarifies what data are to be collected and which type of analyses are going to be executed. This will be elaborated in the next subsection followed by a subsection on privacy and data storage, topics that are relevant when having a research project evaluated by a research ethics committee. Subsequently, the open science practices preregistration and registered reports will be introduced. Both entail the publication of data collection protocols and analysis plans *before* executing them. This is an important step in the open empirical cycle. As can be seen in Figure 1, before registration it is allowed to move back and forth between the first three steps, that is, in the pre-hoc phase there is freedom during the creation of the research design, to reconsider, modify and redesign, whereas after registration, the plans are fixed and should as straightforwardly as possible be executed. The interested reader is referred to Wagenmakers et al. (2018) who distighuish the same two phases in the empirical cycle: the creativity phase (pre-hoc) and

the verification phase (post-hoc).

Planning data collection and analysis

The main issues to consider when planning data collection and analysis, are: the population of interest; the research design, e.g., a randomized controlled trial or a survey (see, for example, Burkholder et al., 2019); whether the data collection is computerized (e.g., web-based questionnaires or lab-based computer tasks) and/or involves human interactions; the data collection protocols (which steps are necessary to collect the data) including all questions, tasks, and computer code (annotated such that it is clear for each section of code what it does); how the variables of interest are measured and coded (see Flake & Fried, 2020 for a paper on questionable measurement practices); which variables (e.g. sumscores) will be computed from the measured variables; which statistical models will be used and which hypotheses with respect to the parameters of these models will be tested; in classical statistics a power analysis (for starters read Cohen, 1992, and Lakens, 2022) and in Bayesian hypothesis evaluation (Hoijtink et al., 2019) an updating plan (for starters read Rouder, 2014 and Elsey, Filmer, & Stemerding, 2021); and, how to deal with missing data (see, for example, van Buuren, 2018) and data exclusion criteria. Each of these issues is relevant when having a research project evaluated by a (medical) research ethics committee and when registering.

While adressing these issues, it can help to test a data collection protocol with procedural pilot participants. These pilot participants are not meant to evaluate the hypotheses of interest or to be included in the final sample, but to test and refine the data collection protocol. It may also be helpful to create or simulate a data set with the desired sample size and containing the variables used in the research project being planned (see, Peikert and Brandmaier, 2021). Creation of such a data set requires a clear research design and clarity about the variables and their scoring (which helps to avoid ambiguity and the necessity of post-hoc decisions later in the empirical cycle). Furthermore, it supports the

creation of analysis scripts, that is, how will data be preprocessed (e.g., missing data analysis and outlier removal) and which analysis will subsequently be executed in what order (again helping to avoid the necessity of posthoc analysis decisions that should be made a priori). An alternative for a simulated data set is to execute a pilot study. As can be seen in Figure 1, while planning data collection and analysis, it is allowed to return to Step 1: Observation or Step 2: Induction and continue from there. This may, for example, happen if the power analysis shows that the required sample sizes are unattainable and research questions and research hypotheses have to be modified such that a simpler statistical model can be used.

Throughout this paper we will refer to Duken et al. (2024) for examples of the implementation of certain open research practices. This is not to say that these examples are the gold standard of the open empirical cycle, but it helps to illustrate open science practices in the context of the empirical cycle. Related to the points above, the paper includes a pilot study in which the general procedure was tested and that provided data with which it was possible write a detailed analysis plan.

Getting approval from the (medical) research ethics committee and formulate data management plan

Before registration, that is, publication of the research plans, it is required to obtain approval from a (medical) ethical research committee. Important in the context of the open empirical cycle are research ethics questions in relation to data management. Therefore in the informed consent form[2] (which should be part of the data collection protocol) it is important to announce the intention to publish the data in line with the FAIR principles (the open science practice connected to Step 4 of the open empirical cycle). Participants should be asked which data may and may not be shared, how, and

[2]https://www.uu.nl/en/research/research-data-management/guides/legal-considerations/how-to-write-an-informed-consent-form

with whom, after the study is completed. Additionally a data management plan[3] has to be created, detailing how secure data storage during and after data collection will be ensured.

Preregistration and registered report

The probability of false positives increases if research plans are modified while collecting and analysing the data (QRPs) in Step 4 of the empirical cycle. Wicherts et al. (2016) provide a taxonomy of different QRPs. Examples in addition to those presented earlier are: excluding one of the treatment groups because the treatment removed is not effective; modifying the criteria for outlier removal; adding and removing covariates, and, increasing the sample size in order to obtain a p-value smaller than .05. De Groot himself already alluded to the open science practice pregistration (see, de Groot et al., 2014, Annotations to Section 1) and Wagenmakers et al. (2012) were among the first to explicitly recommend preregistration, that is, publish research plans in a repository such that the plans cannot be altered, can not be deleted, and will remain publicly available, that is, persistently accesible via a DOI (a digital object identifier[4] which is persistent and uniquely links to the preregistration). Relatively recent Nosek et al. (2019) argued that preregistration is "hard and worthwhile" because it forces researchers to stick to their plans and to highlight deviations in their research report. Peikert et al. (2023) argue that preregistration increases the persuasiveness of research results. An additional benefit is that it may very well be that data collection takes a couple of years, and without a write-up it may be hard to impossible to recollect what the plans were. Preregistration is relevant both for research projects in which data will actually be collected and for research projects using existing data sets (see, Weston et al., 2019, for an elaboration of the latter).

What is a preregistration? One common approach to preregistration is to complete a "preregistration template": a structured document that requires authors to specify their planned analyses in written text. Preregistrations can be published at, and

[3]https://www.uu.nl/en/research/research-data-management/guides/data-management-planning

[4]https://www.doi.org/the-identifier/what-is-a-doi/

templates and examples can be found at, for example, the OSF[5] or ASPREDICTED[6] and additional information is provided in Bakker et al. (2020) and Bosnjak, et al. (2021). Include in the preregistration the DOI to a repository (for example, every preregistration at the OSF is connected to such a repository) where all of data collection protocol and materials, data, analyses, report, and other materials related to the research project will be stored. When writing the preregistration the data collection protocol should be finished and enable anyone to understand precisely how the data will be collected. All the materials used for data collection (e.g., questions asked, questionnaires provided, experimental manipulations, computer code) should be made available. Preferably, these materials are constructed using open software. However, also when using licensed software, the code can often be shared openly, either in its original format or as a simple text file. The more researchers use and share open materials for designing their studies, the easier and faster it will become to modify or create tasks based on publicly existing materials.

Another emerging practice is to forego the use of preregistration templates, and instead, write a full draft of the planned paper, including all planned analyses and placeholder results, using simulated data. The advantages of such a "Preregistration-As-Code" are that authors are not required to write two parallel documents (a preregistration template and a manuscript), and that specifying planned analyses and results in code removes all ambiguity that arises from interpreting plain-text descriptions of planned analyses (Peikert, Van Lissa, & Brandmaier, 2023; Van Lissa, 2022b).

What is a registered report? An alternative for preregistration is a registered report (Chambers, 2019). This is a complete research paper only lacking: a paragraph about data collection, the analyses results, their interpretation, and the discussion. Registered reports can be submitted to an increasing number of journals[7], these are reviewed as usual and only after the paper is accepted for publication the data are

[5]https://help.osf.io/article/330-welcome-to-registrations

[6]https://aspredicted.org/

[7]https://www.cos.io/initiatives/registered-reports

collected and analysed, and the missing sections are added. After the actual execution of the study the journal will verify adherence to the research plans and whether or not they can agree with possible deviations that are highlighted in the research report. If their verification is positive, the report will be published irrespective of the outcome of the analyses. Therefore while preregistrations help to avoid questionable research practices, registered reports additionally help journals to avoid publication bias (Nikolopoulou, 2022) because they decide about accepting or rejecting a paper *before* the data are analysed and the results are known. *An example* of a registered report is Duken et al. (2021).

Step 4: Testing, that is, collect and analyse data

The transition to Step 4 demarcates the pre-hoc from the post-hoc phase of a research project and should therefore be documented and reported clearly, for example through time-stamped preregistrations and reporting of the date of data collection. It is also crucial that any prior exposure to the data is disclosed, including indirect exposure via collaborators or viewing summary statistics. Step 4 comprises three substeps: collect data, analyse data, and exploratory analyses.

Collect data as planned in Step 3. However, unexpected circumstances often occur during data collection (Nosek et al., 2019). Therefore, it is sometimes necessary to adjust the data collection during Step 4. For example, a priori recruitment strategies for participants might not be successful and need to be updated to reach the target sample, or task instructions might not be clear to all participants and need to be improved. Make a research note of such deviations and present and discuss them in the research report.

Analyse data as planned in Step 3 and highlight deviations the plans in the report. Make research notes of deviations from the original plan and present and discuss them in the research report. Interpretations of the analyses should stay true to the theory as formulated in the introduction. At this stage, we should avoid any temptation to re-interpret the theory more expansively to absorb data outside of the scope of the original

claim (theory stretching), or more narrowly so it appears more precisely aligned with the data (post-hoc precision, Frankenhuis, Panchanathan, & Smaldino, 2023). Any updating of the theory should occur, transparently, in Step 5 Evaluation.

Exploratory analyses When executing research plans, it is almost inevitable to get ideas for additional analyses. It may, for example, be of interest to explore the effect of deleting additional outliers or doing additional non-preregistered analyses. The interested reader is also referred to Wigboldus and Dotsch (2016), who "encourage playing with data", Höfler et al. (2022) who discuss the blending of confirmatory and exploratory analyses, and Van Lissa (2022a) who highlights the importance of delineating the scope of exploration and the inferential procedure used to detect effects . Exploratory analyses can be presented in a separate section "Exploratory Analyses" in the research report in which it should be highlighted that exploratory results are tentative and require verification by restarting the open empirical cycle in Step 1 and collecting new data in the ensuing Step 4. This is in line with de Groot who stressed the importance of distinguishing the planned, confirmatory, part of a research project from a possible exploratory part (de Groot et al., 2014). *An example* of exploratory analyses can be found in Duken et al. (2024).

FAIRly publish data and analyses

The open science practice belonging to Step 4 further increases transparency by publishing data, analyses output, and analyses interpretations in a FAIR[8] way such that these can be "F"ound, "A"ccessed, "I"nterpreted, and "R"eused without any obstructions (whether financial or copyright). FAIRly publishing data allows others to criticise and accordingly modify analyses to determine if and how this changes the conclusions in the research paper; the data can be reused in a meta-analysis or research synthesis that combine the results of multiple studies addressing the same research question; and, others can reuse the setup of the analyses if they have a similar research design (van Lissa et al.,

[8]https://www.go-fair.org/fair-principles/

2021). Each "letter" of FAIR will now shortly be elaborated. For elaborations see, for exampe, Wilkinson et al. (2016) and Zondergeld et al. (2020).

The "F" in FAIR denotes Findable. To FAIRly publish data collection protocols, data, and analyses these should be placed in a repository linked to the preregistration via a DOI and coming with it own DOI such that it can easily be referred to and accessed. Such repositories can, for example, easily be created at the Open Science Foundation[9]. Include in the repository a readme file with the contact details of the authors, a description of what can be found in the repository, the link to the preregistration, when finished a link to the research report, and a licence (further discussed below under "R") describing how data and analyses can be reused. Data and analyses can, for example, be published in the form of a JASP[10] file, R users can achieve the same using an R Markdown file[11] or the R package worcs (Van Lissa, et al., 2021) which brings down the learning curve by automating many steps involved in creating FAIR research repositories in R.. All of these are free, that is, can be used without licences or fees.

However, being able to find the files does not yet imply that these can be opened. Therefore, the "A" in FAIR denotes Accessible, that is, the files included in the repository can be opened with FAIR (read open and free) software on various operating systems (Window, MAC, Linux) such that anybody can access the files without encountering (financial or technical) obstacles. The data can be made accessible as soon as these are available. However, if researchers are hesitant to publish data before their report is accepted for publication in a journal, they can first make the data available only for the reviewers and publish FAIRly after their paper has been accepted for publication.

Being able to open the data and analyses files does not yet imply that the content of the files can be understood. The "I" in FAIR denotes Interoperable, that is, those who download data and analyses should be able to interpret them. This implies there should be

[9] https://osf.io/

[10] https://jasp-stats.org/

[11] https://rmarkdown.rstudio.com/

a code book explaining what can be found in the data file and the analyses should be annotated such that it is clear which analyses are executed, which hypotheses are evaluated, and what the interpretation of the results is.

The "R" in FAIR denotes Reusable, which means that data are licensed[12] in such a way that third parties can use them to reproduce the results, include them in a meta analyses, or reuse them to explore new research questions. It is common practice to use creative common licenses[13]. A CC0 license will allow for maximum permissiveness, whereas a CCBY license will ensure that who ever decides to use the materials must cite the authors via a reference. FAIRly publishing data requires careful preparation. It should always be made certain that data are thoroughly anonimized (not traceable to individual persons) and, if researchers are located in Europe, in agreement with the General Data Protection Regulations[14]. Anonimization may to some extend lead to modifications of the data. In these cases it is good practice to highlight how the data were anonimized in a readme file linked to the data. If data contain sensitive information (e.g. about rare diseases, intimate details, or specific demographics) publication is only possible if the research participants have explicitly provided consent for publication *and* reuse by others. When working with personal data however, creative common licenses will not suffice and a data transfer agreement[15] detailing what can and cannot be done with the data must be customized and used when sharing data.

It should be noted[16] that data as facts, cannot be copyrighted, thus what is in fact being licensed is the collected dataset as a whole. The idea is that the data itself refers to the fact depicted by the data such as “the temperature in Cancun on January 1st, 2023 is 27 C” whereas the collection of the dataset is the entire compilation of data stored in a database (file or other) that contains all the temperature measurements of Cancun from

[12]https://www.openaire.eu/how-do-i-license-my-research-data

[13]https://creativecommons.org/about/cclicenses/

[14]https://gdpr-info.eu/

[15]https://utrechtuniversity.github.io/dataprivacyhandbook/data-sharing-reuse.html

[16]https://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=CELEX:31996L0009:EN:HTML

January 1st till January 31st 2023 made by the author. What is copyrighted is not any single datum in the dataset, but the effort placed to compile the data into one file or database with all the necessary metadata and so on.

It may very well be that researchers do not yet want to publish their data because they have additional research ideas and do not want to be scooped. Also in these cases, a big step towards transparency and reproducibility can be made. First of all, only analysis codes and annotated results could be published. With the latter summaries like correlation matrices, means, variances, and sample sizes could be published, that can to some extend be used to corroborate analyses results. Secondly, the part of the data that is or can be thoroughly and completely anonimized could be published. Finally, a synthetic dataset could be generated that allows third parties to verify the computational reproducibility of analyses codes, even if the exact results cannot be reproduced (see, for example, the R worcs package, Van Lissa et al., 2021).

An example of FAIRly published data and analyses from Duken et al. (2024) can be found at OSF[17]

Step 5: Evaluation

This step corresponds to "evaluation" from de Groot's empirical cycle in which the research report is written. It should be based on the theory and expectations formulated in Step 2, the hypotheses, data collection and analyses design formulated in Step 3, and the actual data collection and analysis executed in Step 4. If the research project is preregistered, the preregistration can be the point of departure for writing the research report. In case of a registered report, only the sections on data collection, data analyses, interpretation of the outcomes, and a discussion have to be added. Deviations from the research plans presented in preregistration or registered report, should be highlighted and explained. Strong arguments for highlighting deviations are: transparancy, making

[17] https://osf.io/y7346/

analyses reproducible, and allowing investigation of the impact of deviations on analysis results (Peikert, Van Lissa, & Brandmaier, 2021). In case of major changes to a registration, a research report becomes exploratory in the sense that hypotheses are generated instead of evaluated. It would be "open" to highlight this in the research report. *An example* of highlighting deviations can be found in Duken et al. (2024) who do this in Supplementary Note 9 of their supplementary information[18].

In the discussion of the research report, the actual evaluation takes place. It concerns re-evaluation of the theory based on the empirical results. Research results may support the theory, may require reformulation of the theory, or, may lead to rejection of the theory and formulation of a better theory. Van Lissa et al. (2025) describe an open science workflow for the iterative improvement of FAIR theories. It is important to highlight that although writing the discussion finalizes the research report, it does not end the empirical cycle. Actually, finalization of the research report leads to a re-entry Step 1 of the open empirical cycle with a state of affairs that is updated using the knowledge rendered by the research project that has been executed. It is now time to step back and reconsider the new state of affairs and decide if, what, and why would be valuable further research projects. Exactly this "Return to Step 1" is what makes the open empirical cycle cyclic in the sense that it offers the option to continuously accumulate and update knowledge. *An example* of explicitly returning to Step 1 can be found dispersed in the Discussion of Duken et al. (2024). There the theories under investigation are primarily interpreted in the context of the confirmatory analysis results. However, our data also inspired new ideas, which lead to exploratory analyses and the consideration of a different theory that we did not include a priori. Returning to step 1 means that the ideas derived from our data and exploratory analyses can be the starting point for new confirmatory research projects. Our shared materials and data may further help other researchers to built on our project. Rather than starting form scratch, they can use our task or they can use our data and

[18]https://doi.org/10.1038/s41467-024-45203-4

analysis scripts for their own studies with our data.

The open science practice associated with evaluation is publication of research results in the form of an open access publication. Publishing open access is an important feature of open science. Being able to unobstructedly obtain everything (also the report) related to a research project will enable anyone to: benefit from the research, reuse (parts) of the research, and to engage in a fully informed discussion. Furthermore, it avoids that null findings disappear in the "file drawer" (Rosenthal, 1979) and ensures that anybody (also non-academic audiences) can access the publication. Releasing a preprint (Bourne et al., 2017) of the research report may increase attention and citations (Fu and Hughey, 2019; McKiernan et al., 2016). This can be done by adding it to the research repository, or, for example, at psyarxiv[19] where it comes with its own DOI. This "green open access" ensures that anyone can obtain and read the report without costs being involved. Of course researchers also want to publish their report in a journal. On openaccess.nl[20] arguments in favor of and information about open access publication can be found. In addition to green open access there are "gold open access" which requires authors to pay for their publication and "diamond or platinum open access" (see, for example, the journal Meta Psychology[21]) where both publication and access of scientific papers is completely free.

In case of preregistered research, reviewers will read in the introduction of the research report where research plans have been registered, and that deviations from the plans have been highlighted. This will enable them to compare plans to what has actually been executed. If data and analyses are published FAIRly, reviewers can make their review even more thorough by reproducing analyses and while doing so, evaluate the adequacy of the codes used. When a completed registered report is submitted to a journal it is reviewed before and after adding sections on data collection, data analyses, interpretation of the outcomes, and a discussion. Stated otherwise, when using open science practices reviewers

[19]https://psyarxiv.com

[20]https://www.openaccess.nl/en

[21]https://open.lnu.se/index.php/metapsychology/index

are enabled to provide a more thorough evaluation of the research report, which should increase the trustworthyness of the research executed. *An example* of reviews of both the registered report and the final report can be found in the "Peer Review File"[22].

Discussion

The open empirical cycle should be viewed as a tool that supports transparent scholarly workflows in the pursuit of cumulative knowledge acquisition. It is a reminder of the most important research steps that have to be considered, decisions that have to be made, and issues that have to be considered. The structured approach to science advocated by the open empirical cycle may actually save time because it protects from having to double back or construct ad-hoc reparations because earlier in the research process something was forgotten. Additionally, Frankenhuis and Nettle (2018) argue that open science practices are not a burden, but psychologically liberating. They reward research quality (which is under a reserearchers control) instead of (significant) research outcomes (which is not under a researchers control), and "foster creativity because they cultivate an open and flexible mind-set, create a more collaborative and constructive climate, and generate more accurate information and make it more accessible (p. 439)".

However, being open may lead to feelings of vulnerability. To phrase this differently, "what if someone detects a mistake?", "will such a mistake influence the future of (early) career scientists?" One of the advantages of being open is that editors and reviewers can not only read the research report, but also verify what was actually done. Consequently, it is more likely that mistakes will be detected, and if any remain, it escaped not only the authors attention, but also the attention of editor and reviewers. Furthermore, one of the purposes of being open is to allow others to scrutinize research work, compare preregistration to report, express their support or disagreement, provide alternative analyses and interpretations, and, possibly, detect mistakes. It is important to note that

[22]https://doi.org/10.1038/s41467-024-45203-4

criticism should be directed at the research, not the researchers. It is equally important to note that if others scrutinize and evaluate a research project this can be seen as a compliment because apparantly the research is interesting and worthwhile to them.

It is important to highlight that open science is not an all-or-nothing game. It may very well be necessary to employ certain open science practices while not employing others. Stated otherwise, research projects are located on a continuum from open to closed. Registered reports with publicly available materials, code, and data are probably the highest standard of open science at the moment, but even employing only one of the open science practices will already increase the transparancy and trustworthiness of research.

The benefit of using open science practices are threefold. First, having an open track record is an advantage because funding agencies increasingly require open publications and open data from the projects they fund. Second, doing good quality transparant research is valued higher than having many publications. The interested reader is also referred to the Declaration of Research Assessment[23] which is signed by a steadily increasing number of individuals and organisations. Adhering (if only partly) to the open empirical cycle, is a preparation for this shift in perspective. Third, also important, using open science practices may increase *your own trust* in *your own research* (the interested reader is referred to Rohrer et al. (2021) that contains statements of researchers that did lose the trust in their own research).

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Author Contributions

Herbert Hoijtink conceptualized the article. All authors were involved in writing, reviewing, and editing the article. Herbert Hoijtink and Sascha Duken took the lead in rewriting the article after receiving reviews.

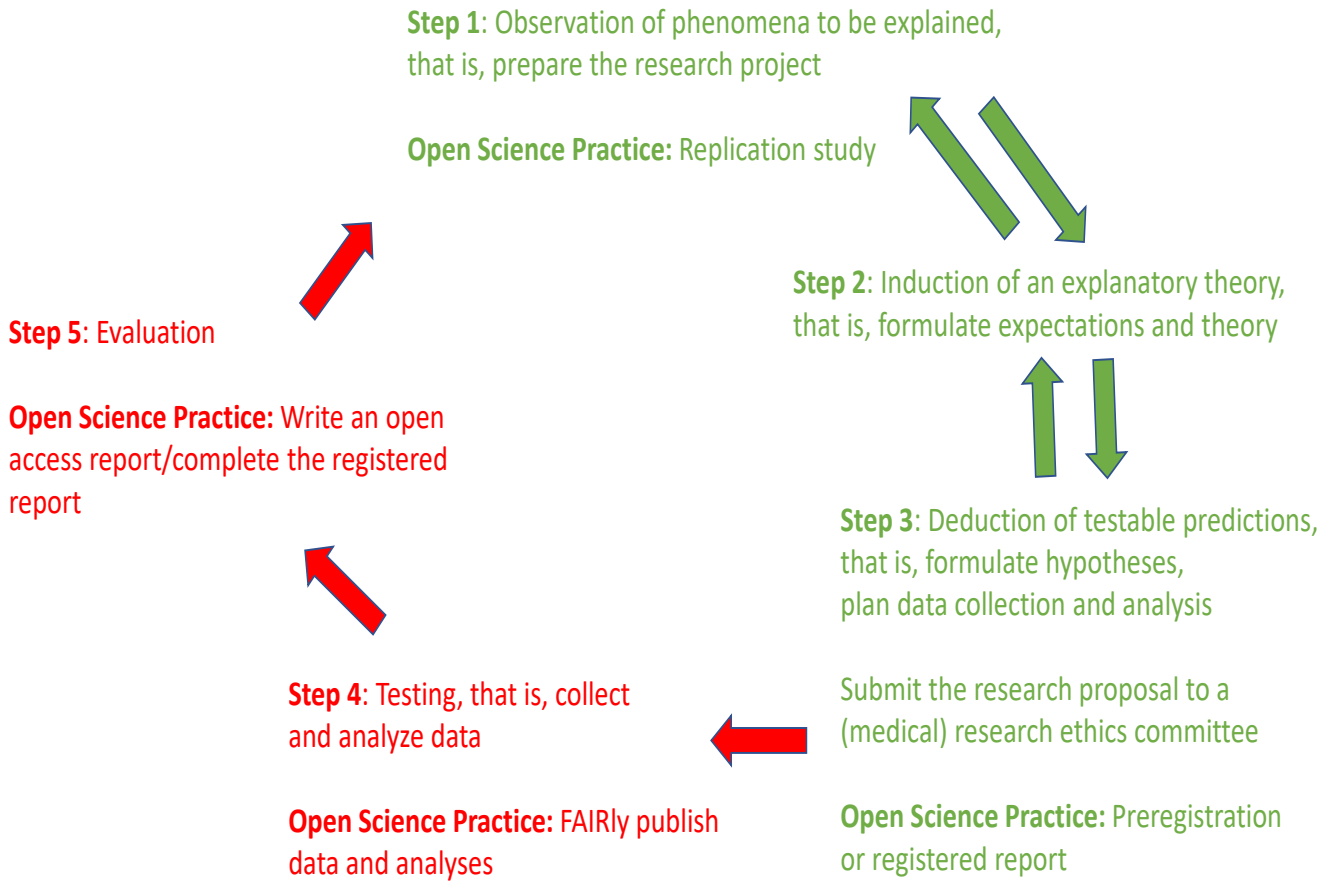


Figure 1. The open empirical cycle, that is, de Groot's (1961) empirical cycle integrated with four open science practices. The prehoc (before collecting the data) phase is in green, the posthoc phase in red. In the prehoc phase earlier steps can be revisited as indicated by the arrows.